Question 4: Numerical Study of the Burgers' Equation

Travelling waves, CNLF finite differences, and Chebyshev collocation

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MH7007 Topics in Scientific Computation I

April 16, 2026

Question 4: Problem Statement

We study the viscous Burgers' equation

$$u_t + uu_x = \varepsilon u_{xx}, \qquad x \in \mathbb{R}, \quad t > 0, \quad \varepsilon > 0. \tag{1}$$

The assignment asks us to:

1. verify the exact travelling-wave solution;
2. formulate and test a Crank–Nicolson–Leap-Frog (CNLF) finite-difference method;
3. solve the interval problem on $(-1, 1)$ using CNLF in time and Chebyshev collocation in space.

Conservative form will be central:

$$u_t + \frac{1}{2}(u^2)_x = \varepsilon u_{xx}. \tag{2}$$

Part 1: Travelling-Wave Ansatz

The proposed solution is

$$u(x,t) = \omega\left(1 - \tanh\left[\frac{\omega}{2\varepsilon}(x - \omega t - x_0)\right]\right). \tag{3}$$

Let

$$y = x - \omega t - x_0, \qquad u(x,t) = w(y). \tag{4}$$

Then

$$u_t = -\omega w', \qquad u_x = w', \qquad u_{xx} = w''. \tag{5}$$

Substitution into Burgers' equation gives

$$-\omega w' + ww' = \varepsilon w'' \quad \Longleftrightarrow \quad -\omega w' + \left(\frac{w^2}{2}\right)' = \varepsilon w''. \tag{6}$$

First Integral and Separable ODE

Integrating once in y,

$$-\omega w + \frac{w^2}{2} = \varepsilon w' + C. \tag{7}$$

For a travelling front with

$$w(-\infty) = 2\omega, \qquad w(+\infty) = 0, \qquad w'(\pm\infty) = 0, \tag{8}$$

we obtain $C = 0$. Hence

$$\varepsilon w' = -\omega w + \frac{w^2}{2} = \frac{w(w - 2\omega)}{2}. \tag{9}$$

Separating variables,

$$\frac{dw}{w(w - 2\omega)} = \frac{dy}{2\varepsilon}. \tag{10}$$

Solving the Travelling-Wave ODE

Using partial fractions,

$$\frac{1}{2\omega}\int\left(\frac{1}{w-2\omega}-\frac{1}{w}\right)dw=\frac{y}{2\varepsilon}. \tag{11}$$

Absorbing the integration constant into x_0 gives

$$w(y)=\frac{2\omega}{1+\exp(\omega y/\varepsilon)}. \tag{12}$$

Since

$$\frac{2}{1+e^A}=1-\tanh\left(\frac{A}{2}\right), \tag{13}$$

we get

$$\boxed{u(x,t)=\omega\left(1-\tanh\left[\frac{\omega}{2\varepsilon}(x-\omega t-x_0)\right]\right).} \tag{14}$$

Direct Verification

Define

$$\eta = \frac{\omega}{2\varepsilon}(x - \omega t - x_0), \qquad u = \omega(1 - \tanh \eta). \tag{15}$$

Since $\eta_x = \omega/(2\varepsilon)$ and $\eta_t = -\omega^2/(2\varepsilon)$,

$$u_x = -\frac{\omega^2}{2\varepsilon} \operatorname{sech}^2 \eta, \tag{16}$$

$$u_t = \frac{\omega^3}{2\varepsilon} \operatorname{sech}^2 \eta, \tag{17}$$

$$u_{xx} = \frac{\omega^3}{2\varepsilon^2} \operatorname{sech}^2 \eta \tanh \eta. \tag{18}$$

Thus

$$u_t + u u_x = \frac{\omega^3}{2\varepsilon} \operatorname{sech}^2 \eta \tanh \eta = \varepsilon u_{xx}. \tag{19}$$

Part 2: Conservative Form and CNLF Grid

Use

$$u_t + \frac{1}{2}(u^2)_x = \varepsilon u_{xx}. \tag{20}$$

Let

$$x_m = x_L + mh, \qquad t_n = nk, \qquad V_m^n \approx u(x_m, t_n). \tag{21}$$

Discretize:

$$u_t(x_m, t_n) \approx \frac{V_m^{n+1} - V_m^{n-1}}{2k}, \tag{22}$$

$$\frac{1}{2}(u^2)_x(x_m, t_n) \approx \frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{4h}, \tag{23}$$

$$u_{xx}(x_m, t_n) \approx \frac{\delta_x^2 V_m^{n+1} + \delta_x^2 V_m^{n-1}}{2h^2}. \tag{24}$$

CNLF Scheme

Here

$$\delta_x^2 V_m^n = V_{m+1}^n - 2V_m^n + V_{m-1}^n. \tag{25}$$

Substituting into the PDE gives

$$\frac{V_m^{n+1} - V_m^{n-1}}{2k} + \frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{4h} = \frac{\varepsilon}{2h^2}\left(\delta_x^2 V_m^{n+1} + \delta_x^2 V_m^{n-1}\right). \tag{26}$$

With $\mu = \varepsilon k/h^2$,

$$\boxed{V_m^{n+1} - \mu\delta_x^2 V_m^{n+1} = V_m^{n-1} + \mu\delta_x^2 V_m^{n-1} - \frac{k}{2h}\left((V_{m+1}^n)^2 - (V_{m-1}^n)^2\right).} \tag{27}$$

Why This Is a Linear Solve

At time level $n+1$, the nonlinear term is already known because it is evaluated at level n.

Structure

- Leap-frog in time for u_t.
- Explicit centered conservative flux for $(u^2)_x/2$.
- Crank–Nicolson averaging for the viscous term.

Therefore the unknown vector V^{n+1} solves a tridiagonal linear system:

$$\left(I - \mu\delta_x^2\right) V^{n+1} = \text{known right-hand side.} \tag{28}$$

Since CNLF is a two-step method, it requires V^0 and V^1. For the travelling-wave convergence test, V^1 is taken from the exact solution.

Local Truncation Error

Taylor expansions for a smooth solution give

$$\frac{u_m^{n+1} - u_m^{n-1}}{2k} = u_t + \frac{k^2}{6} u_{ttt} + O(k^4), \tag{29}$$

$$\frac{(u_{m+1}^n)^2 - (u_{m-1}^n)^2}{4h} = \frac{1}{2}(u^2)_x + \frac{h^2}{12}(u^2)_{xxx} + O(h^4), \tag{30}$$

$$\frac{\delta_x^2 u_m^{n+1} + \delta_x^2 u_m^{n-1}}{2h^2} = u_{xx} + \frac{k^2}{2} u_{xxtt} + \frac{h^2}{12} u_{xxxx} + O(k^4 + k^2 h^2 + h^4). \tag{31}$$

Hence

$$\tau_m^n = O(k^2 + h^2). \tag{32}$$

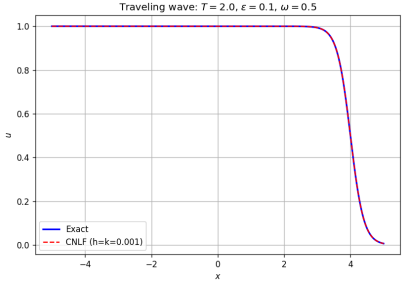
Travelling-Wave Numerical Test

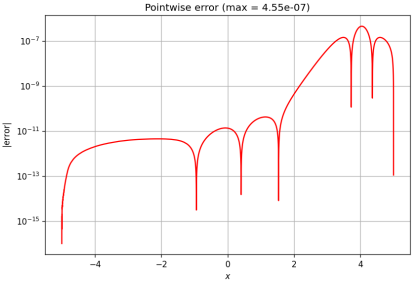
Parameters:

$$\omega = 0.5, \quad \varepsilon = 0.1, \quad x_0 = 3, \quad -5 \le x \le 5, \quad T = 2, \quad h = k = 10^{-3}. \tag{33}$$

Notebook result:

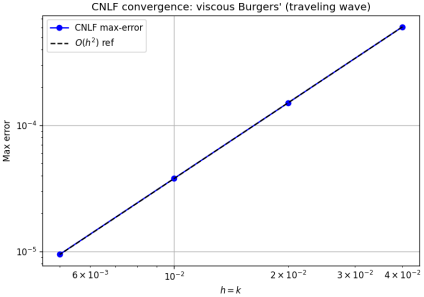
$$\|e(T)\|_\infty = 4.548 \times 10^{-7}. \tag{34}$$





Coupled Convergence: $h = k$

$h = k$	$\|e(T)\|_\infty$	rate
0.0400	6.056×10^{-4}	–
0.0200	1.513×10^{-4}	2.001
0.0100	3.790×10^{-5}	1.997
0.0050	9.485×10^{-6}	1.998



The observed slope is second order.

Separate Time and Space Convergence

Temporal convergence
$h = 2.5 \times 10^{-4}$

k	$\|e\|_2$	rate
$8.0 \cdot 10^{-3}$	$3.663 \cdot 10^{-6}$	–
$4.0 \cdot 10^{-3}$	$9.125 \cdot 10^{-7}$	2.005
$2.0 \cdot 10^{-3}$	$2.250 \cdot 10^{-7}$	2.020
$1.0 \cdot 10^{-3}$	$5.316 \cdot 10^{-8}$	2.081

Spatial convergence
$k = 10^{-4}$

h	$\|e\|_2$	rate
$4.0 \cdot 10^{-2}$	$1.584 \cdot 10^{-4}$	–
$2.0 \cdot 10^{-2}$	$3.963 \cdot 10^{-5}$	1.999
$1.0 \cdot 10^{-2}$	$9.911 \cdot 10^{-6}$	1.999
$5.0 \cdot 10^{-3}$	$2.478 \cdot 10^{-6}$	2.000

Both experiments confirm $O(k^2)$ and $O(h^2)$ behaviour.

Part 3: Interval Problem

Now solve Burgers' equation on $(-1, 1)$:

$$u_t + uu_x = \varepsilon u_{xx}, \qquad -1 < x < 1, \quad t > 0, \tag{35}$$

with

$$u(\pm 1, t) = 0, \qquad u(x, 0) = -\sin(\pi x), \qquad \varepsilon = 0.02, \qquad T = 0.995. \tag{36}$$

Spectral collocation choice

Fourier collocation is natural for periodic domains. For a nonperiodic interval, the corresponding standard spectral collocation method is Chebyshev collocation.

Chebyshev–Lobatto Grid

Use the Chebyshev–Lobatto points

$$x_j = \cos\left(\frac{j\pi}{N}\right), \qquad j = 0, \ldots, N. \tag{37}$$

These nodes cluster near $x = \pm 1$, helping resolve boundary layers and steep gradients.
For nodal values $U_j(t) \approx u(x_j, t)$,

$$p_N'(x_i) \approx \sum_{j=0}^{N} D_{ij} U_j. \tag{38}$$

For $i \neq j$,

$$D_{ij} = \frac{c_i}{c_j} \frac{(-1)^{i+j}}{x_i - x_j}, \qquad c_0 = c_N = 2, \quad c_j = 1. \tag{39}$$

The diagonal is chosen by $D_{ii} = -\sum_{j \neq i} D_{ij}$, and $D^{(2)} = D^2$.

Chebyshev Semi-Discrete Burgers System

In conservative form,

$$\dot{U} + \frac{1}{2} D(U \odot U) = \varepsilon D^2 U, \tag{40}$$

where $\odot$ denotes componentwise multiplication.
Impose the Dirichlet conditions:

$$U_0 = U_N = 0. \tag{41}$$

On the interior nodes $I = \{1, \ldots, N-1\}$, the CNLF update is

$$(I - \varepsilon k D_{II}^2) U_I^{n+1} = U_I^{n-1} + \varepsilon k (D^2 U^{n-1})_I - \frac{k}{2} \left(D((U^n)^2) \right)_I . \tag{42}$$

Chebyshev Startup Step

Since CNLF is a two-step method, compute U^1 from a nonlinear Crank–Nicolson startup:

$$U^1 - U^0 + \frac{k}{4} D\left((U^1)^2 + (U^0)^2\right) - \frac{\varepsilon k}{2} D^2(U^1 + U^0) = 0. \tag{43}$$

This nonlinear system is solved by Newton's method on the interior nodes.
Parameters used:

$$N = 96, \qquad k = 10^{-3}, \qquad \varepsilon = 0.02, \qquad T = 0.995. \tag{44}$$

Chebyshev-Collocation Result

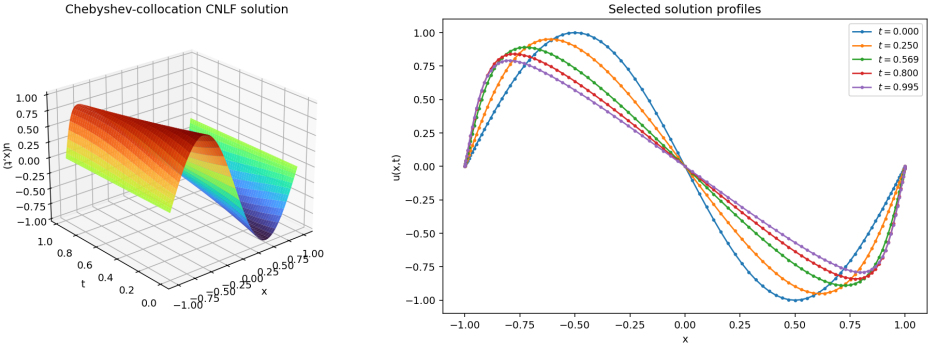
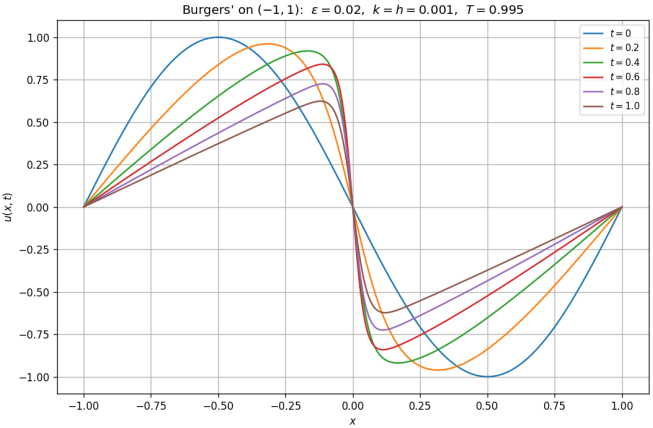


Figure 1: Chebyshev-collocation CNLF solution and selected profiles.

Finite-Difference Reference Profiles



The Chebyshev method captures the same steepening and viscous smoothing with a spectral spatial discretisation.

Takeaways

- The tanh travelling wave is obtained by a travelling-wave reduction and satisfies Burgers' equation exactly.
- Conservative form gives a clean finite-difference approximation of the nonlinear term.
- The CNLF method combines leap-frog time stepping, explicit centered flux, and Crank–Nicolson diffusion.
- Taylor expansion gives local truncation error $O(k^2 + h^2)$.
- Numerical tests confirm second-order accuracy.
- Chebyshev collocation is the standard spectral alternative for the nonperiodic interval problem.